Білет № 1

1. HEFT/CPOP scheduling algorithms.
2. Non-relational models: network, hierarchical, document, graph. CAP theorem.
3. Page-oriented storage engines. B-tree.
4. Normal forms. ACID concept.
5. Log-structured tables. Iceberg table format.

Білет № 2

1. DAGs. Representation of programs as DAGs. Scheduling of programs on DAGs.
2. Replication types (single/multi-leader, leaderless).
3. Parallelization. Multithreading. Multiprocessing.
4. HEFT/CPOP scheduling algorithms.
5. Distributed systems. Features. Sharing types (memory/disk/nothing).

Білет № 3

1. Python GIS. Raster and vector data. GeoDataFrame. xarray and rioxarray datatypes. K-nearest neighbor search. Spatial joins.
2. Normal forms. ACID concept.
3. Consistency. Partitioning.
4. Relational data model. Data independence. Relational algebra definitions/operations.
5. RUM conjecture. LSM-trees vs B-trees.

Білет № 4

1. DAGs. Representation of programs as DAGs. Scheduling of programs on DAGs.
2. Log-structured tables. Iceberg table format.
3. Non-relational models: network, hierarchical, document, graph. CAP theorem.
4. Integrity: relational, domain, atomic.
5. Data formats: textual vs binary. Google Dremel. Apache Parquet.

Білет № 5

1. Dask internals: scheduling types, dask graph, collections. Blocked algorithms.
2. Definition of Big Data. Three Vs of Big Data. Complexity - definition. Accidental vs incidental.
3. Consistency. Partitioning.
4. Integrity: relational, domain, atomic.
5. Distributed systems. Features. Sharing types (memory/disk/nothing).

Білет № 6

1. OLTP vs OLAP. Data warehousing. Data lakes. Lakehouse concept.
2. Python GIS. Raster and vector data. GeoDataFrame. xarray and rioxarray datatypes. K-nearest neighbor search. Spatial joins.
3. Query languages comparison: SQL, Cypher, Datalog.
4. Parallelization. Multithreading. Multiprocessing.
5. Relational data model. Data independence. Relational algebra definitions/operations.

Білет № 7

1. Data formats: textual vs binary. Google Dremel. Apache Parquet.
2. HEFT/CPOP scheduling algorithms.
3. Log-structured tables. Iceberg table format.
4. RUM conjecture. LSM-trees vs B-trees.
5. Definition of Big Data. Three Vs of Big Data. Complexity - definition. Accidental vs incidental.

Білет № 8

1. DAGs. Representation of programs as DAGs. Scheduling of programs on DAGs.
2. Query languages comparison: SQL, Cypher, Datalog.
3. Parallelization. Multithreading. Multiprocessing.
4. Log-structured tables. Iceberg table format.
5. Dask internals: scheduling types, dask graph, collections. Blocked algorithms.

Білет № 9

1. Log-structured tables. Iceberg table format.
2. Page-oriented storage engines. B-tree.
3. Normal forms. ACID concept.
4. Query languages comparison: SQL, Cypher, Datalog.
5. Distributed systems. Features. Sharing types (memory/disk/nothing).

Білет № 10

1. Integrity: relational, domain, atomic.
2. Query languages comparison: SQL, Cypher, Datalog.
3. Replication types (single/multi-leader, leaderless).
4. Normal forms. ACID concept.
5. Definition of Big Data. Three Vs of Big Data. Complexity - definition. Accidental vs incidental.

Білет № 11

1. RUM conjecture. LSM-trees vs B-trees.
2. Query languages comparison: SQL, Cypher, Datalog.
3. Consistency. Partitioning.
4. Relational data model. Data independence. Relational algebra definitions/operations.
5. Log-structured storage engines. SSTable, LSM-tree.

Білет № 12

1. Distributed systems. Features. Sharing types (memory/disk/nothing).
2. Relational data model. Data independence. Relational algebra definitions/operations.
3. Parallelization. Multithreading. Multiprocessing.
4. Dask internals: scheduling types, dask graph, collections. Blocked algorithms.
5. Data formats: textual vs binary. Google Dremel. Apache Parquet.